

Artifact Subspace Reconstruction (ASR)

EEG Preprocessing Method — Summary & Validation

1) What ASR does (short & clear)

Artifact Subspace Reconstruction (ASR) is an EEG cleaning method designed to suppress short, high-amplitude artifacts (such as eye blinks and muscle bursts) while preserving the underlying neural signal. It works by analyzing the EEG in small time windows, detecting abnormal activity, and reconstructing only the corrupted part.

2) Why ASR is useful in EEG

EEG recordings often contain transient artifacts that can dominate the signal and degrade feature extraction or ICA. ASR reduces these extreme segments automatically, improving data quality and stability for downstream processing.

3) Implementation note (MNE usage)

In this project, MNE is used as a reliable EEG container (RawArray) to handle multi-channel time-series data consistently. The ASR logic itself is implemented window-by-window using robust statistics and PCA-based reconstruction.

4) How we verified ASR works correctly

We validated ASR using both quantitative checks (numbers) and qualitative checks (visual inspection).

Validation Check	Expected Result	Observed Result
Blink artifact peak-to-peak (90–170)	Decrease after ASR	7.93 → 5.09
Muscle burst peak-to-peak (240–330)	Decrease after ASR	10.53 → 5.09
RMS change outside artifact windows	Small (near 0)	0.018

Interpretation: Outside the artifact windows, the ASR modification is minimal (RMS difference = 0.018), meaning ASR mainly targets corrupted segments without distorting stable EEG.

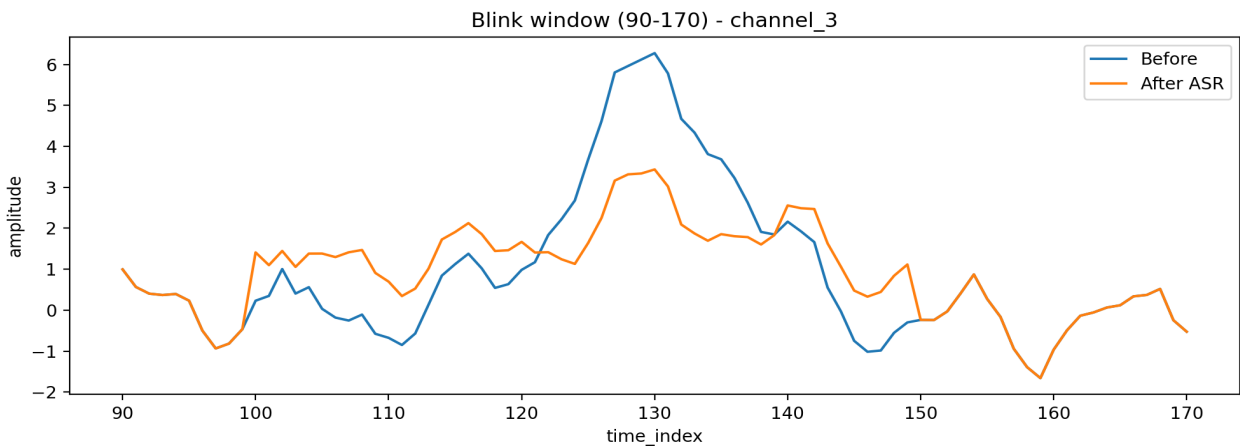


Figure 1 — Blink window (Channel_3, samples 90–170). The high-amplitude blink peak is reduced after ASR while the surrounding EEG waveform remains stable.

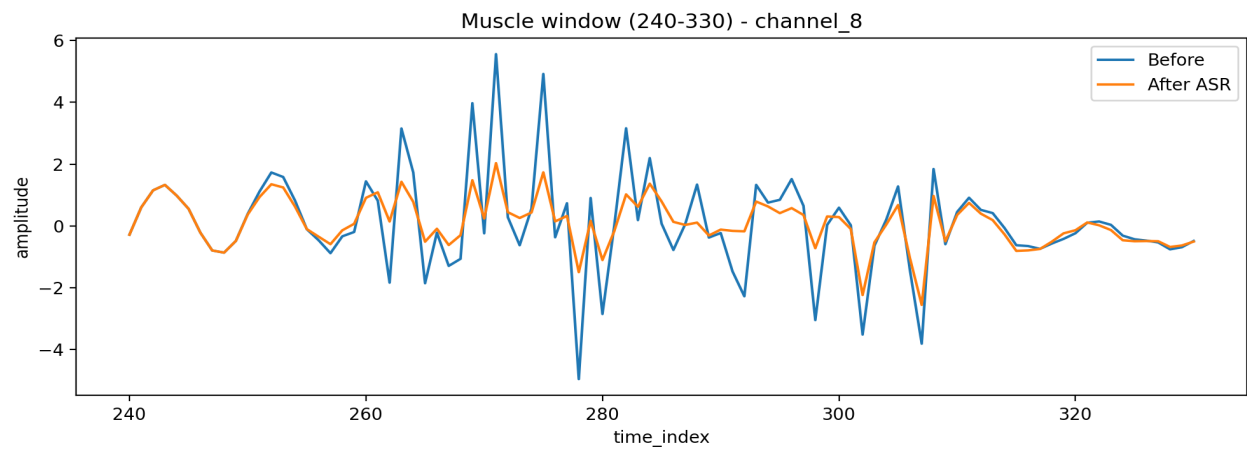


Figure 2 — Muscle burst window (Channel_8, samples 240–330). High-frequency muscle activity is suppressed after ASR, indicating correct artifact attenuation.

Key takeaway

ASR is considered correct when transient, high-amplitude artifacts are reduced in affected windows, while the overall EEG waveform remains stable outside artifact periods.